higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## MATHEMATICS N6

## 29 JULY 2019

**This marking guideline consists of 20 pages.**

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**NOTE: This paper is marked out of 200 and divided by 2 to get a mark out of 100.**

**QUESTION 1**

1.1 $z = x^2 + 2xy + y^2$

$\frac{\partial z}{\partial x} = 2x + 2y$ ✓ $\quad \frac{\partial z}{\partial y} = 2x + 2y$ ✓

$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = x(2x + 2y) + y(2x + 2y)$ ✓

$= (2x + 2y)(x + y)$ ✓

$= 2(x + y)(x + y)$ ✓

$= 2(x + y)^2$

$= 2(x^2 + 2xy + y^2)$ ✓

$= 2z$

Alternative

$z = x^2 + 2xy + y^2$

$= (x + y)^2$ ✓ ✓ ✓ ✓

$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = x.2(x + y).1 + y.2(x + y).1$

$= 2(x + y)(x + y)$ ✓

$= 2(x + y)$ ✓

$= 2z$

(6)

1.2 $x = \sqrt{t} = t^{\frac{1}{2}} \quad y = \frac{1}{\sqrt{t}} = t^{-\frac{1}{2}}$

✓ ✓

$\frac{dx}{dt} = \frac{1}{2}t^{-\frac{1}{2}} \quad \frac{dy}{dt} = -\frac{1}{2}t^{-\frac{3}{2}}$

$\frac{dy}{dx} = \frac{-\frac{1}{2}t^{-\frac{3}{2}}}{\frac{1}{2}t^{-\frac{1}{2}}} = -t^{-1}$ ✓ ✓

$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx} = t^{-2}\frac{1}{\frac{1}{2}t^{-\frac{1}{2}}} = t^{-2}2t^{\frac{1}{2}} = 2t^{-\frac{3}{2}}$ ✓ ✓

$x = \sqrt{t} \qquad y = \frac{1}{\sqrt{t}}$

$y = \frac{1}{x}$ ✓

$\frac{dy}{dx} = -\frac{1}{x^2} = -x^{-2}$ ✓ ✓

$\frac{d^2y}{dx^2} = 2x^{-3}$ ✓

$= 2\left(t^{\frac{1}{2}}\right)^{-3} = 2t^{-\frac{3}{2}}$ ✓ ✓

(6)
**[12]**

## QUESTION 2

2.1 $\int y dx = \int x^2 e^{3x} dx \qquad f(x) = x^2 \qquad g'(x) = e^{3x}$

$$= x^2 \frac{e^{3x}}{3} - \int 2x \frac{e^{3x}}{3} dx$$

$$= x^2 \frac{e^{3x}}{3} - \left[ 2x \frac{e^{3x}}{9} - \int 2 \frac{e^{3x}}{9} dx \right]$$

$$= x^2 \frac{e^{3x}}{3} - 2x \frac{e^{3x}}{9} + \frac{2}{27} e^{3x} + c \qquad \checkmark\checkmark\checkmark\checkmark\checkmark\checkmark$$

Alternative

| $f(x)$ | $g'(x)$ | |
|---|---|---|
| $x^2$ | $e^{3x}$ | |
| | $\frac{e^{3x}}{3}$ | ✓ |
| $2x$ | | |
| | $\frac{e^{3x}}{9}$ | ✓ |
| $2$ | | |
| | $\frac{e^{3x}}{27}$ | ✓ |
| $0$ | | |

$$\boxed{x^2 \frac{e^{3x}}{3} - 2x \frac{e^{3x}}{9} + 2 \frac{e^{3x}}{27} + c} \quad \checkmark \checkmark \checkmark$$

(6)

2.2 $\int \cos^5 \frac{x}{5} dx = \int \cos^4 \frac{x}{5} \cos \frac{x}{5} dx \quad \checkmark \quad \checkmark$

$$= \int \left( \cos^2 \frac{x}{5} \right) \cos \frac{x}{5} dx$$

$$= \int \left( 1 - \sin^2 \frac{x}{5} \right)^2 \cos \frac{x}{5} dx \quad \checkmark \qquad u = \sin \frac{x}{5} \qquad \frac{du}{dx} = \frac{1}{5} \cos \frac{x}{5}$$

$$= \int \left( 1 - u^2 \right)^2 \cos \frac{x}{5} \frac{5du}{\cos \frac{x}{5}} \quad \checkmark \qquad dx = \frac{5du}{\cos \frac{x}{5}}$$

$$= 5 \int \left( 1 - u^2 \right)^2 du \quad \checkmark$$

$$= 5 \int \left( 1 - 2u^2 + u^4 \right) du \quad \checkmark$$

$$= 5 \left[ u - 2 \frac{u^3}{3} + \frac{u^5}{5} \right] + c \quad \checkmark$$

$$= 5\left[\sin\frac{x}{5} - 2\frac{\sin^3\frac{x}{5}}{3} + \frac{\sin^5\frac{x}{5}}{5}\right] + c \quad \checkmark$$

$$= 5\sin\frac{x}{5} - \frac{10}{3}\sin^3\frac{x}{5} + \sin^5\frac{x}{5} + c \qquad (8)$$

2.3 $\int \tan^3 x \sec x dx$

$= \int \tan^2 x \sec x \tan x dx$ ✓

$= \int (\sec^2 x - 1)\sec x \tan x dx$ ✓

$= \int (u^2 - 1)du$

$= \frac{u^3}{3} - u + c$ ✓

$= \frac{\sec^3 x}{3} - \sec x + c$ ✓ ✓ ✓

| | | |
|---|---|---|
| $u = \sec x$ | $\frac{du}{dx} = \sec x \tan x$ | *kkk* |
| | $dx = \frac{du}{\sec x \tan x}$ | |

Alternative

$\int \tan^3 x \sec x dx \qquad u = \tan x$

$= \int u^3 \sec x \frac{du}{\sec^2 x}$ ✓ $\qquad \frac{du}{dx} = \sec^2 x \qquad dx = \frac{du}{\sec^2 x}$

$= \int u^3 \frac{du}{\sec x}$ ✓

$= \int u^3 \frac{du}{\sqrt{1+u^2}} \qquad v = 1 + u^2 \qquad \Rightarrow \frac{dv}{du} = 2u \Rightarrow du = \frac{dv}{2u}$

$= \int \frac{u^3}{v^{\frac{1}{2}}} \frac{dv}{2u} = \frac{1}{2}\int u^2 v^{-\frac{1}{2}} dv$ ✓

$= \frac{1}{2}\int (v-1)v^{-\frac{1}{2}} dv \quad = \frac{1}{2}\int v^{\frac{1}{2}} - v^{-\frac{1}{2}} dv$ ✓

$= \frac{1}{2}\left[\frac{v^{\frac{3}{2}}}{\frac{3}{2}} - \frac{v^{\frac{1}{2}}}{\frac{1}{2}}\right] = \frac{1}{3}v^{\frac{3}{2}} - v^{\frac{1}{2}}$ ✓

$= \frac{1}{3}\sec^3 x - \sec x + c$ ✓

Alternative

$\int \tan^3 x \sec x dx$

$\int \frac{\sin^3 x}{\cos^3 x}\frac{1}{\cos x}dx$

$\int \frac{\sin^3 x}{\cos^4 x}dx$ ✓

$= \int \frac{\sin^2 x \sin x}{\cos^4 x}dx$

$= \int \frac{1-\cos^2 x \sin x}{\cos^4 x}dx$ ✓ $\quad u = \cos x \quad \frac{du}{dx} = -\sin x$

$= \int \frac{1-u^2 \sin x}{u^4}\frac{du}{-\sin x}$

$dx = \frac{du}{-\sin x}$

$= -\int \frac{1-u^2}{u^4}du$ ✓

$= -\int u^{-4} - u^{-2}du$ ✓

$= -\left[\frac{u^{-3}}{-3} - \frac{u^{-1}}{-1}\right] + c$ ✓

$= -\left[\frac{(\cos x)^{-3}}{-3} - \frac{(\cos x)^{-1}}{-1}\right] + c \ = \frac{\sec^3 x}{3} - \sec x + c$ ✓

(6)

2.4

$\int \frac{1}{9-4x-x^2}dx$

$= \int \frac{1}{13-(x+2)^2}dx$ ✓

$= \frac{1}{2\sqrt{13}} \ln \frac{\sqrt{13}+(x+2)}{\sqrt{13}-(x+2)} + c$ ✓ ✓ ✓

$9-4x-x^2$

$= -\left[x^2+4x+4-9-4\right]$ ✓

$= -\left[(x+2)^2 - 13\right]$ ✓ ✓

$= 13-(x+2)^2$ ✓

or $\frac{1}{7,211}$ *or* $0,139 \ln \frac{\sqrt{13}+(x+2)}{\sqrt{13}-(x+2)} + c$

or $-\int \frac{1}{(x+2)^2-13}dx$

$= -\frac{1}{2\sqrt{13}} \ln \frac{(x+2)-\sqrt{13}}{(x+2)+\sqrt{13}} + c$

$ax^2+bx+c$

$= \frac{4ac-b^2}{4a} + a\left(x+\frac{b}{2a}\right)^2$

$9-4x-x^2$

$= \frac{4(-1)9-(4)^2}{4(-1)} - \left(x+\frac{-4}{2(-1)}\right)^2$

$= \frac{-36-16}{-4} - (x+2)^2$

$= 13-(x+2)^2$

(8)

2.5 $\int \frac{1}{ab}\tan^{-1}\frac{bx}{a}dx \qquad f(x)=\tan^{-1}\frac{bx}{a}$ and $g'(x)=\frac{1}{ab}$

✓ ✓

$$=\frac{1}{ab}x\tan^{-1}\frac{bx}{a}-\int\frac{1}{ab}x\frac{\frac{b}{a}}{1+\frac{b^2x^2}{a^2}}dx \quad ✓$$

$$=\frac{1}{ab}x\tan^{-1}\frac{bx}{a}-\int\frac{1}{ab}x\frac{ba}{a^2+b^2x^2}dx \quad ✓$$

$$=\frac{1}{ab}x\tan^{-1}\frac{bx}{a}-\int\frac{x}{a^2+b^2x^2}dx \quad ✓$$

$$=\frac{1}{ab}x\tan^{-1}\frac{bx}{a}-\frac{1}{2b^2}\int\frac{2b^2x}{a^2+b^2x^2}dx \quad ✓$$

$$==\frac{1}{ab}x\tan^{-1}\frac{bx}{a}-\frac{1}{2b^2}\ln\left(a^2+b^2x^2\right)+c \quad ✓ \quad ✓$$

$u=a^2+b^2x^2$

$\frac{du}{dx}=2b^2x$

$dx=\frac{du}{2b^2x}$

Alternative

$$\int \frac{1}{ab}\tan^{-1}\frac{bx}{a}dx$$

$$u = \tan^{-1}\frac{bx}{a} \qquad \frac{bx}{a} = \tan u$$

$$\frac{b}{a} = \sec^2 u\frac{du}{dx} \qquad dx = \frac{a}{b}\sec^2 udu$$

$$\therefore \int \frac{1}{ab}\tan^{-1}\frac{bx}{a}dx$$

$$= \int \frac{1}{ab}u\frac{a}{b}\sec^2 udu$$

$$= \frac{1}{b^2}\int u\sec^2 udu \qquad f(u) = u \qquad g'(u) = \sec^2 u$$

$$= \frac{1}{b^2}\left[u\tan u - \int \tan udu\right]$$

$$\frac{1}{b^2}\left[u\tan u - \ln(\sec u)\right]$$

$$\frac{1}{b^2}\left[u\tan u - \ln(1+\tan^2 u)^{\frac{1}{2}}\right]$$

$$\frac{1}{b^2}\left[u\tan u - \frac{1}{2}\ln(1+\tan^2 u)\right]$$

$$\frac{1}{b^2}\left[\tan^{-1}\frac{bx}{a}\left(\frac{bx}{a}\right) - \frac{1}{2}\ln(1+\frac{b^2x^2}{a^2})\right] + c$$

$$= \frac{1}{b^2}\left[\left(\frac{bx}{a}\right)\tan^{-1}\frac{bx}{a} - \frac{1}{2}\ln\frac{a^2+b^2x^2}{a^2}\right] + c$$

$$= \frac{1}{b^2}\left[\left(\frac{bx}{a}\right)\tan^{-1}\frac{bx}{a} - \frac{1}{2}\ln\left(a^2+b^2x^2\right) + \frac{1}{2}\ln a^2\right] + c$$

$$= \frac{1}{ab}x\tan^{-1}\frac{bx}{a} - \frac{1}{2b^2}\ln\left(a^2+b^2x^2\right) + \frac{1}{2b^2}\ln a^2 + c$$

$$= \frac{1}{ab}x\tan^{-1}\frac{bx}{a} - \frac{1}{2b^2}\ln\left(a^2+b^2x^2\right) + K$$

where K is a constant $= \frac{1}{2b^2}\ln a^2 + c$

(8)
**[36]**

## QUESTION 3

3.1

$$\int \frac{8x^2-2x+3}{2x^3-2x^2+x-1}dx$$

$2x^3-2x^2+x-1=2x^2(x-1)+(x-1)$ ✓

$=(x-1)(2x^2+1)$ ✓

$\frac{8x^2-2x+3}{2x^3-2x^2+x-1}=\frac{8x^2-2x+3}{(x-1)(2x^2+1)}=\frac{A}{x-1}+\frac{Bx+C}{2x^2+1}$ ✓ ✓

$8x^2-2x+3=A(2x^2+1)+(Bx+C)(x-1)$ ✓

$x=1 \quad 8-2+3=A(2+1) \qquad \therefore A=3$ ✓

$8x^2-2x+3=(2Ax^2+A)+(Bx^2+Cx-Bx-C)$ ✓

$2A+B=8 \qquad \therefore B=2$ ✓

$C-B=-2 \qquad \therefore C=0$ ✓

$$\int \frac{8x^2-2x+3}{2x^3-2x^2+x-1}dx=\int \frac{3}{x-1}dx+\int \frac{2x}{2x^2+1}dx$$ ✓

$$=3\ln(x-1)+\frac{1}{2}\ln(2x^2+1)+c$$ ✓ ✓

Alternative

$$\int \frac{8x^2-2x+3}{2x^3-2x^2+x-1}dx$$ ✓

$2x^3-2x^2+x-1=2x^2(x-1)+(x-1)$ ✓

$=(x-1)(2x^2+1)$ ✓

and $8x^2-2x+3=6x^2+3+2x^2-2x$

$=3\left(2x^2+1\right)+2x(x-1)$ ✓

$$\therefore \quad \frac{8x^2-2x+3}{2x^3-2x^2+x-1}=\frac{3\left(2x^2+1\right)+2x(x-1)}{(x-1)(2x^2+1)}$$ ✓ ✓

$$=\frac{3\left(2x^2+1\right)}{(x-1)(2x^2+1)}+\frac{2x(x-1)}{(x-1)(2x^2+1)}$$ ✓

$$=\frac{3}{x-1}+\frac{2x}{2x^2+1}$$ ✓ ✓

$$\therefore \int \frac{8x^2-2x+3}{2x^3-2x^2+x-1}dx=\int \frac{3}{x-1}dx+\int \frac{2x}{2x^2+1}dx$$ ✓

$$=3\ln(x-1)+\frac{1}{2}\ln(2x^2+1)+c$$ ✓ ✓

(12)

3.2 $\dfrac{(3x+2)(2x-3)}{(3x+2)^2-(2x-3)^2}=\dfrac{6x^2-5x-6}{5x^2+24x-5}$ ✓✓

$$¿\frac{6}{5}+\frac{\frac{-169}{5}x}{(5x-1)(x+5)} \quad \textit{using long division method}$$ ✓✓

$$=1{,}2+\frac{-33{,}8x}{(5x-1)(x+5)} \quad \textit{where}$$

$$\frac{\frac{-169}{5}x}{(5x-1)(x+5)}=\frac{A}{(5x-1)}+\frac{B}{x+5}$$ ✓

$$-\frac{169}{5}x=A(x+5)+B(5x-1)$$ ✓

$x=-5 \quad B=-6{,}5=-\dfrac{13}{2}$ ✓

$x=\dfrac{1}{5} \quad A=-1{,}3=-\dfrac{13}{10}$ ✓

$$\int\frac{(3x+2)(2x-3)}{(3x+2)^2-(2x-3)^2}dx=\int\frac{6}{5}dx-\int\frac{13}{10(5x-1)}dx-\int\frac{13}{2(x+5)}dx$$ ✓

$$¿\frac{6}{5}x-\frac{13}{50}\ln(5x-1)-\frac{13}{2}\ln(x+5)+C$$ ✓✓✓

(12)
**[24]**

## QUESTION 4

4.1

$$\frac{dy}{dx} = \tan x - y\cot x$$

$$\frac{dy}{dx} + y\cot x = \tan x \quad \checkmark$$

$$e^{\int p dx} = e^{\int \cot x dx} \quad \checkmark$$

$$= e^{\ln(\sin x)} \quad \checkmark$$

$$= \sin x \quad \checkmark$$

$$\int Q e^{\int p dx} = \int \tan x \sin x dx \quad \checkmark$$

$$= \int \frac{\sin x}{\cos x} \sin x dx$$

$$= \int \frac{\sin^2 x}{\cos x} dx \quad \checkmark$$

$$= \int \frac{1-\cos^2 x}{\cos x} dx \quad \checkmark$$

$$= \int \frac{1}{\cos x} - \frac{\cos^2 x}{\cos x} dx \quad \checkmark$$

$$= \int \sec x - \cos x dx \quad \checkmark$$

$$= \ln(\sec x + \tan x) - \sin x + c \quad \checkmark \quad \checkmark$$

$$\therefore y\sin x = \ln(\sec x + \tan x) - \sin x + c \quad \checkmark$$

(12)

4.2

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 2e^{2x}$$

$$r^2 - 2r + 2 = 0 \quad \checkmark$$

$$r = \frac{2 \pm \sqrt{4 - 4.1.2}}{2} \quad \checkmark$$

$$= \frac{2 \pm \sqrt{-4}}{2} \quad \checkmark$$

$$= \frac{2 \pm 2i}{2} = 1 \pm i \quad \checkmark$$

$$y_c = e^x \left[A\cos x + B\sin x\right] \quad \checkmark$$

$y = Ce^{2x}$ ✓

$\frac{dy}{dx} = 2Ce^{2x}$ ✓

$\frac{d^2y}{dx^2} = 4Ce^{2x}$ ✓

$\therefore 4Ce^{2x} - 2\left(2Ce^{2x}\right) + 2Ce^{2x} = 2e^{2x}$ ✓

$C = 1$ ✓

$y_p = e^{2x}$

$y = y_c + y_p$ ✓

$y = e^x\left[A\cos x + B\sin x\right] + e^{2x}$ ✓

**OR**

$y = 2Ce^{2x}$

$\frac{dy}{dx} = 4Ce^{2x}$

$\frac{d^2y}{dx^2} = 8Ce^{2x}$

$\therefore 8Ce^{2x} - 2\left(4Ce^{2x}\right) + 2(2Ce^{2x}) = 2e^{2x}$

$4Ce^{2x} = 2e^{2x} \quad \Rightarrow 4C = 2 \quad \therefore C = \frac{1}{2}$

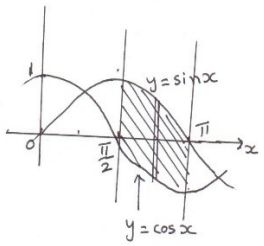
$y_p = 2Ce^{2x}$

$y_p = 2\left(\frac{1}{2}\right)e^{2x} = e^{2x}$

(12)
**[24]**

## QUESTION 5

5.1 5.1.1




✓ shape of each graph (2 x 1)
✓ x intercepts $\frac{\pi}{2}$ and $\pi$ (2 x 1)
✓ strip
✓ area

(6)

5.1.2

$Area = \int_a^b y_1 - y_2 dx$ ✓

$= \int_{\frac{\pi}{2}}^{\pi} \sin x - \cos x dx$

$= \left[-\cos x - \sin x\right]_{\frac{\pi}{2}}^{\pi}$ ✓

$= \left[-\cos\pi - \sin\pi - \{-\cos\frac{\pi}{2} - \sin\frac{\pi}{2}\}\right]$ ✓

or $-\left[\cos\pi + \sin\pi - \{\cos\frac{\pi}{2} + \sin\frac{\pi}{2}\}\right]$

$= 2\text{units}^2$

(8)

5.1.3

$$A_{m-y} = \int rdA \quad \checkmark$$

$$= \int_{\frac{\pi}{2}}^{\pi} x(\sin x - \cos x)dx \quad \checkmark \checkmark \quad \checkmark \; f(x) = x \qquad g'(x) = \sin x - \cos x$$

$$= \left[ x(-\cos x - \sin x) - \int (-\cos x - \sin x)dx \right]_{\frac{\pi}{2}}^{\pi} \quad \checkmark \quad \checkmark$$

$$= \quad + \left[ x(\cos x + \sin x) - \int (\cos x + \sin x)dx \right]_{\frac{\pi}{2}}^{\pi}$$

$$= -\left[ x(\cos x + \sin x) - (\sin x - \cos x) \right]_{\frac{\pi}{2}}^{\pi} \quad \checkmark$$

$$= -\left[ \pi(\cos \pi + \sin \pi) - (\sin \pi - \cos \pi) - \left\{ \frac{\pi}{2}(\cos \frac{\pi}{2} + \sin \frac{\pi}{2}) - (\sin \frac{\pi}{2} - \cos \frac{\pi}{2}) \right\} \right] \quad \checkmark\checkmark$$

$$= -\left[ \pi(-1+0) - (0--1) - \left\{ \frac{\pi}{2}(0+1) - (1-0) \right\} \right]$$

$$= -\left[ -\pi - 1 - \frac{\pi}{2} + 1 \right]$$

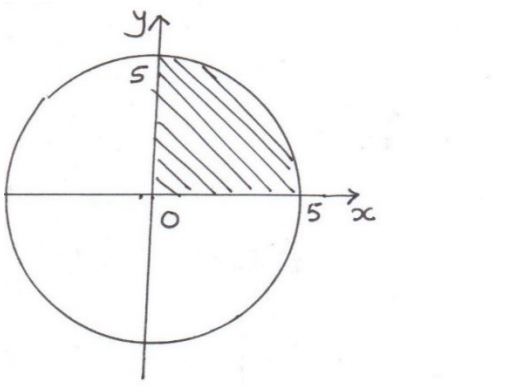
$$= \frac{3\pi}{2} \quad \text{or} \quad 4,712 \ \text{units}^3 \quad \checkmark$$

$$\checkmark$$

$$\bar{x} = \frac{A_{m-y}}{A} = \frac{4,712}{2} = 2,356 \ units \quad \checkmark$$

(12)

5.2 5.2.1




✓ strip (vertical or horizontal used in calculation)
✓ area
✓ graph (shape)
✓ intercept

(4)

5.2.2

$$V_x = \pi\int_a^b y_1^2 - y_2^2\,dx \quad \checkmark$$

$$= \pi\int_0^5 25 - x^2\,dx \quad \checkmark \quad \checkmark$$

$$= \pi\left[25x - \frac{x^3}{3}\right]_0^5 \quad \checkmark$$

$$= \pi\left[25(5) - \frac{(5)^3}{3}\right] \quad \checkmark$$

$$= \frac{250}{3}\pi \quad \text{or} \quad 261,800 \text{ units}^3 \quad \checkmark$$

(6)

5.2.3

$$V_{m-y} = \pi\int_a^b x\left(y_1^2 - y_2^2\right)dx \quad \checkmark$$

$$= \pi\int_0^5 x\left(25 - x^2\right)dx \quad \checkmark \quad \checkmark$$

$$= \pi\int_0^5 25x - x^3\,dx \quad \checkmark$$

$$= \pi\left[\frac{25}{2}x^2 - \frac{x^4}{4}\right]_0^5 \quad \checkmark$$

$$= \frac{625}{4}\pi = 156,25\pi = 490,874\text{units}^4 \quad \checkmark$$

$$\bar{x} = \frac{261,800}{490,874} = \frac{15}{8} = 1,875 \text{ units} \quad \checkmark \quad \checkmark$$

$$V_{m-y} = \int_a^b r\,dv$$

$$= \int_a^b \frac{x_1 + x_2}{2} 2\pi y\left(x_1 - x_2\right)dy \quad \checkmark$$

$$= \pi\int_a^b y\left(x_1^2 - x_2^2\right)dy \quad \checkmark \quad \checkmark$$

$$= \pi\int_0^5 y\left(25 - y^2\right)dy \quad \checkmark$$

$$= \pi\int_a^5 \left(25y - y^3\right)dy \quad \checkmark$$

$$= \pi\left[\frac{25y^2}{2} - \frac{y^4}{4}\right]_0^5 \quad \checkmark$$

$$= \pi\left[\frac{25(5)^2}{2} - \frac{y(5)^4}{4} - \{0\}\right] \quad \checkmark$$

$$= \frac{625}{4}\pi \quad \text{or} \quad 490,874 \text{ units}^4 \quad \checkmark$$

$$\bar{x} = \frac{261,800}{490,874} = \frac{15}{8} = 1,875 \text{ units}$$

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Using horizontal strip

$$V_x = 2\pi\int_a^b y\left(x_1 - x_2\right)dy \quad \checkmark$$

$$= 2\pi\int_0^5 y\sqrt{25 - y^2}\,dy \quad \checkmark \checkmark$$

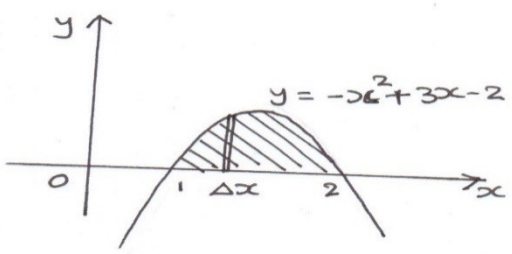
$$= -\pi\int_0^5 -2y\left(25 - y^2\right)^{\frac{1}{2}}dy$$

$$= -\frac{2}{3}\pi\left[\left(25 - y^2\right)^{\frac{3}{2}}\right]_0^5 \quad \checkmark \quad \checkmark$$

$$= -\frac{2}{3}\pi\left[0 - 25^{\frac{3}{2}}\right] = \frac{250}{3}\pi \quad \checkmark$$

$$= 83,333\pi = 261,800\text{units}^3$$

5.3 5.3.1




✓ graph
✓ x-intercepts
✓ area
✓ strip (4)

5.3.2

$$A = \int_a^b y_1 - y_2\,dx \quad \checkmark$$

$$= \int_1^2 -x^2 + 3x - 2\,dx \quad \checkmark$$

$$= \left[-\frac{x^3}{3} + \frac{3x^2}{2} - 2x\right]_1^2 \quad \checkmark \quad \checkmark$$

$$= \left[-\frac{(2)^3}{3} + \frac{3(2)^2}{2} - 2(2) - \left\{-\frac{1}{3} + \frac{3}{2} - 2\right\}\right] \quad \checkmark$$

$$= \frac{1}{6} \text{ or } 0,167 \text{ units}^2 \quad \checkmark$$

(6)

5.3.3

$$I_y = \int_a^b r^2\,dA \quad \checkmark$$

$$= \int_1^2 x^2(-x^2 + 3x - 2)\,dx \quad \checkmark$$

$$\int_1^2 (-x^4 + 3x^3 - 2x^2)\,dx \quad \checkmark$$

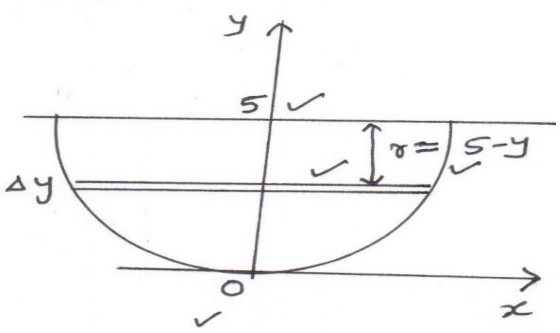
$$= \left[ -\frac{x^5}{5} + \frac{3x^4}{4} - \frac{2x^3}{3} \right]_1^2 \quad \checkmark$$

$$\left[ -\frac{(2)^5}{5} + \frac{3(2)^4}{4} - \frac{2(2)^3}{3} - \left\{ -\frac{1}{5} + \frac{3}{4} - \frac{2}{3} \right\} \right] \quad \checkmark$$

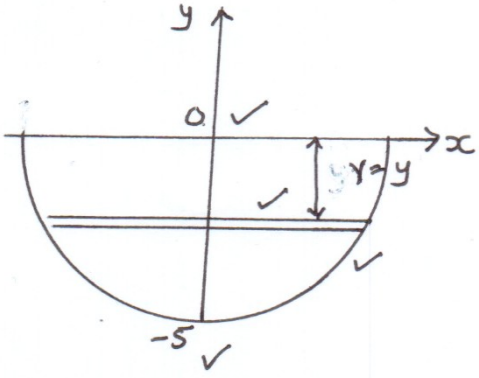
$$= \frac{23}{60} \quad or 0,383 \quad \checkmark = \frac{0,383}{0,167} A = 2,295A \quad \checkmark \checkmark$$

(8)

5.4 5.4.1




Alternative (with x-axis at the water level)




(4)

5.4.2 $x^2 + (y-5)^2 = 25$

$x = \sqrt{25-(y-5)^2}$ ✓

first moment $= \int_a^b rdA$ ✓

$= \int_0^5 (5-y)\,2\sqrt{25-(y-5)^2}\,dy$ ✓ ✓ ✓ $u = 5-y \quad dy = -du$

$= -2\int_5^0 u\sqrt{25-u^2}\,du$ ✓ $y = 0 \quad u = 5$

$= \frac{2}{3}\left[\left(25-u^2\right)^{\frac{3}{2}}\right]_5^0$ ✓ $y = 5 \quad u = 0$

$= \frac{2}{3}\left[25^{\frac{3}{2}} - 0\right]$ ✓ $= \frac{250}{3} = 83,333\ m^3$ ✓

$= \left[\frac{\left(25-u^2\right)^{\frac{3}{2}}}{\frac{3}{2}}\right]_5^0$

$= \frac{2}{3}\left[\left(25-u^2\right)^{\frac{3}{2}}\right]_5^0$

$= \frac{2}{3}\left[(25)^{\frac{3}{2}} - \{0\}\right]$

$= \frac{250}{3}$ or $83,333m^3$

5.4.3 $= y = \frac{245,437}{83,333} = 2,945m$ ✓ ✓

Alternative

5.4.2 First moment of area $= \int_a^b rdA$ ✓

$= \int_a^b y2xdy$ ✓✓

But $x^2 + y^2 = 25 \Rightarrow x = \sqrt{25-y^2}$ ✓✓

$= \int_{-5}^0 y2\sqrt{25-y^2}\,dy$ ✓✓

$= -\int_{-5}^0 -y2\left(25-y^2\right)^{\frac{1}{2}}\,dy$ ✓

$= -\left[\frac{\left(25-y^2\right)^{\frac{3}{2}}}{\frac{3}{2}}\right]_{-5}^0$ ✓

$= -\frac{2}{3}\left[\left(25-y^2\right)^{\frac{3}{2}}\right]_{-5}^0$ ✓

(12)

$$= -\frac{2}{3}\left[(25)^{\frac{3}{2}} - \left\{25 - (-5)^2\right\}^{\frac{3}{2}}\right] \quad \checkmark$$

$$= -\frac{2}{3}(25)^{\frac{3}{2}} = -83\frac{1}{3} \quad \text{or} -\frac{250}{3}m^3 \quad \checkmark \checkmark$$

5.4.3 $$\overset{=}{y} = \frac{245,437m^4}{-83,333m^3} = -2,945m \quad \checkmark \quad \checkmark$$

(2)
**[80]**

## QUESTION 6

6.1 $2y = x^2$

$\frac{dy}{dx} = x$ ✓

$\left[\frac{dy}{dx}\right]^2 = x^2$ ✓

$1+\left[\frac{dy}{dx}\right]^2 = 1+x^2$ ✓

$S = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dy$ ✓

$= \int_2^4 \sqrt{1+x^2}\,dy$ ✓ ✓

$= \left[\frac{x}{2}\sqrt{1+x^2} + \frac{1}{2}\ln\left(x+\sqrt{1+x^2}\right)\right]_2^4$ ✓ ✓

$\left[\frac{4}{2}\sqrt{1+4^2} + \frac{1}{2}\ln\left(4+\sqrt{1+4^2}\right) - \left\{\frac{2}{2}\sqrt{1+2^2} + \frac{1}{2}\ln\left(2+\sqrt{1+2^2}\right)\right\}\right]$ ✓ ✓

$= \left[2\sqrt{17} + \frac{1}{2}\ln\left(4+\sqrt{17}\right) - \left\{\sqrt{5} + \frac{1}{2}\ln\left(2+\sqrt{5}\right)\right\}\right]$

$= 6,336$ units ✓ ✓

Alternative

(12)

$2y = x^2$

$x = \sqrt{2}y^{\frac{1}{2}} \qquad or \sqrt{2}\sqrt{y}$

$\frac{dx}{dy} = \sqrt{2}\frac{1}{2}y^{-\frac{1}{2}}$ ✓

$\left[\frac{dx}{dy}\right]^2 = \frac{1}{2y}$ ✓

$1+\left[\frac{dx}{dy}\right]^2 = 1+\frac{1}{2y}$ ✓

$= \frac{2y+1}{2y}$

$S = \int_a^b \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$ ✓

$= \int_2^8 \sqrt{\frac{2y+1}{2y}}\,dy$

$\int_2^8 \frac{\sqrt{2y+1}}{\sqrt{2y}}dy$ ✓

$= \int_2^4 \frac{\sqrt{1+u^2}}{u}udu$ ✓

$= \int_2^4 \sqrt{1+u^2}\,du$ ✓

$= \left[\frac{u}{2}\sqrt{1+u^2}+\frac{1}{2}\ln\left(u+\sqrt{1+u^2}\right)\right]_2^4$ ✓ ✓

$\left[\frac{4}{2}\sqrt{1+4^2}+\frac{1}{2}\ln\left(4+\sqrt{1+4^2}\right)-\left\{\frac{2}{2}\sqrt{1+2^2}+\frac{1}{2}\ln\left(2+\sqrt{1+2^2}\right)\right\}\right]$ ✓

$= \left[2\sqrt{17}+\frac{1}{2}\ln\left(4+\sqrt{17}\right)-\left\{\sqrt{5}+\frac{1}{2}\ln\left(2+\sqrt{5}\right)\right\}\right]$

$= 6,336$ units ✓ ✓

6.2

$$x = \frac{1}{9}y^2$$

$$\frac{dx}{dy} = \frac{2}{9}y \quad \checkmark$$

$$\left[\frac{dx}{dy}\right]^2 = \left(\frac{2}{9}y\right)^2 \quad \checkmark$$

$$1+\left[\frac{dx}{dy}\right]^2 = 1+\left(\frac{2}{9}y\right)^2 = 1+\frac{4y^2}{81} = \frac{81+4y^2}{81} \quad \checkmark \ \checkmark$$

$$A_x = 2\pi\int_0^6 y\frac{\sqrt{81+4y^2}}{9}dy \quad \checkmark \ \checkmark$$

$$= \frac{2}{9}\pi\int_{81}^{225} y\,u^{\frac{1}{2}}\frac{du}{8y}$$

$$= \frac{2}{9}\frac{1}{8}\pi\int_{81}^{225} u^{\frac{1}{2}}du$$

$$= \frac{1}{36}\pi\left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right]_{81}^{225}$$

$$= \frac{1}{36}\frac{2}{3}\pi\left[u^{\frac{3}{2}}\right]_{81}^{225}$$

$$= \frac{1}{54}\pi\left[225^{\frac{3}{2}} - 81^{\frac{3}{2}}\right]$$

$$= 49\pi \text{ or } 153{,}938 \text{ units}^2$$

Alternative

$x = \frac{1}{9}y^2$

$y^2 = 9x \quad \checkmark \Rightarrow y = 3\sqrt{x}$

$2y\frac{dy}{dx} = 9$

$\frac{dy}{dx} = \frac{9}{2y} = \frac{9}{2\left(3\sqrt{x}\right)} = \frac{9}{6\sqrt{x}} \quad = \frac{3}{2\sqrt{x}} \quad \checkmark \quad \checkmark$

$\left[\frac{dy}{dx}\right]^2 = \left(\frac{3}{2\sqrt{x}}\right)^2 \quad \checkmark$

$1 + \left[\frac{dy}{dx}\right]^2 = 1 + \left(\frac{3}{2\sqrt{x}}\right)^2 = 1 + \frac{9}{4x} = \frac{4x+9}{4x} \quad \checkmark$

$A_x = 2\pi\int_0^4 y\sqrt{1 + \left[\frac{dy}{dx}\right]^2}\,dx \quad \checkmark \; y = 0 \quad x = 0$

$= 2\pi\int_0^4 y\sqrt{\frac{4x+9}{4x}}dx \quad \checkmark \qquad y = 6 \quad x = \frac{1}{9}y^2 = \frac{1}{9}(6)^2 = 4 =$

$2\pi\int_0^4 3\sqrt{x}\,\frac{\sqrt{4x+9}}{2\sqrt{x}}dx \quad \checkmark$

$= 3\pi\int_0^4 \sqrt{4x+9}dx$

$= 3\pi\frac{1}{4}\left[\frac{(4x+9)^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^4 \quad \checkmark$

$\frac{1}{2}\pi\left[(4x+9)^{\frac{3}{2}}\right]_0^4$

$= \frac{1}{2}\pi\left[(4(4)+9)^{\frac{3}{2}} - (0+9)^{\frac{3}{2}}\right] \quad \checkmark$

$\frac{1}{2}\pi\left[(25)^{\frac{3}{2}} - (9)^{\frac{3}{2}}\right]$

$= 49\pi \;\text{ or }\; 153{,}938\text{units}^2 \quad \checkmark \; \checkmark$ (12)

**[24]**

**TOTAL** = 200 ÷ 2**:** **100**

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